Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Aston-Jones, Gary Ph.D.		
Project Title:	Stress, Performance and Locus Coeruleus		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gaj@mail.med.upenn.edu	**Fax:**	FY 215-573-5201
PI Organization Type:	UNIVERSITY	**Phone:**	215-573-5200
Organization Name:	University of Pennsylvania		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	02/01/2001	**End Date:**	08/31/2004
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Druhan, Jonathan		
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NBPF.002.08		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Establish tasks for rats that measure sustained attention. Record and manipulate activity of locus coeruleus neurons during task performance to establish role of this system in performance. Determine effects of different stressors on actiivty of locus coeruleus neurons and performance. Pharmacologically antagonize stress-inputs to locus coeruleus and test whether this attenuates effects of stress on performance.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Determine the role of the noradrenergic locus coeruleus system in performance and attention, and in the effects of stress on performance. Identify pharmacologic manipulations to attenuate effects of stress on LC activity, and thereby on performance. This work will be of benefit to situations on earth where humans must attend and perform under stress, e.g., radar monitoring, air traffic controllers, and many man-machine interface applications.
Task Progress:	No progress provided this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)